

3 Method

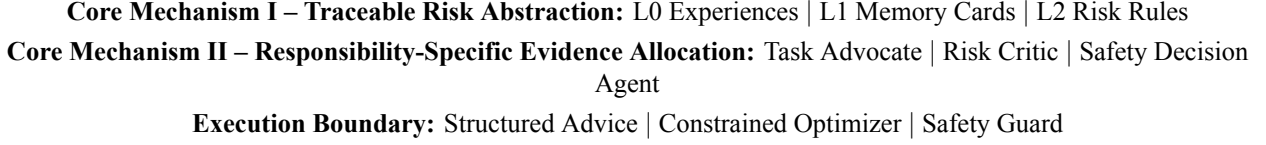


Fig. 1: Method overview. Two coupled mechanisms transform interaction history into traceable risk abstractions and allocate them by decision responsibility; deterministic modules retain execution authority.

3.1 System Overview

We propose a risk-experience-driven multi-role framework whose key novelty is the coupling of *traceable risk abstraction* and *responsibility-specific evidence allocation*. Given interaction history $\mathcal{H}_t$, the first mechanism converts complete interactions into compact case-level cards and cross-case rules while preserving the evidence supporting each abstraction. Given the current planning input $\mathcal{X}_t = (g_t, r_t, e_t, \mathcal{A}_t)$, the second mechanism selects a different evidence bundle for each decision responsibility. Consequently, long-term experience is compressed according to its risk structure and then exposed selectively according to how it will be used in deliberation.

The Task Advocate, Risk Critic, and Safety Decision Agent interpret their assigned evidence and return structured recommendations with cited evidence IDs. A constrained optimizer then selects a semantic action from the admissible set $\mathcal{A}_t$, and an independent Safety Guard validates it before execution. The agents therefore provide evidence-grounded semantics but never executable control. Section 3.2 defines the traceable abstraction mechanism; Section 3.3 defines responsibility-specific allocation and its integration with decision making.

3.2 Traceable Hierarchical Risk Memory

Our first mechanism constructs a hierarchy in which each level has a distinct decision function. L0 experiences $\mathcal{E}_t$ preserve complete interactions, L1 cards $\mathcal{M}_t$ expose the risk structure of individual cases, and L2 rules $\mathcal{Q}_t$ encode patterns supported across cases. The defining property is traceable abstraction: every compression step retains stable links to its inputs. The links determine rule support, validate decision citations, and identify the cases affected by later contradictory evidence. Thus, the hierarchy reduces decision-time context while keeping each abstraction empirically verifiable and updatable.

3.2.1 L0: Embodied Risk Experiences

L0 represents an interaction as E_i , which records its task and scene conditions, action A_i , outcome O_i , failure cause, risk type, severity $\sigma_i \in \{1, \dots, 5\}$, and trajectory provenance. An immutable `experience_id` links the record to sensor logs, controls, or human labels.

To compare heterogeneous observations on a common risk-oriented scale, we encode E_i as $z_i = [1 - \bar{c}_i, 1 - \bar{d}_i, 1 - p_i, 1 - b_i, h_i]$. Here $\bar{c}_i$ and $\bar{d}_i$ are normalized clearance and obstacle distance, p_i is observation confidence, b_i is battery ratio, and h_i contains discrete robot, obstacle, and task conditions. The complements make larger values indicate stronger risk factors; L0 retains the full trajectory for audit.

3.2.2 L1: Risk Memory Cards

L1 compresses each experience into a risk memory card. The normalized severity $\sigma_i/5$ represents consequence magnitude, the outcome weight $\omega(O_i)$ assigns 1, 0.8, or 0.05 to failure, aborted, or success, and $\gamma_i \in [0, 1]$ discounts evidence with low record quality. Their product $q_i = (\sigma_i/5)\omega(O_i)\gamma_i$ is a relative evidence score, not a collision probability. Let F_i denote the failure cause, B_i the mitigation and stopping conditions, and P_i the source record containing the memory ID, experience ID, and source span.

Substituting z_i and q_i into the compression map derives the L1 card:

$$\begin{aligned} M_i = C(E_i) &= \text{Pack}(z_i, A_i, O_i, F_i, B_i, q_i, P_i) \\ &= \text{Pack}([1 - \bar{c}_i, 1 - \bar{d}_i, 1 - p_i, 1 - b_i, h_i], \\ &\quad A_i, O_i, F_i, B_i, \frac{\sigma_i}{5}\omega(O_i)\gamma_i, P_i). \end{aligned} \quad (1)$$

Equation (1) shows how the risk state, outcome evidence, and provenance of an L0 experience jointly form its traceable L1 card.

3.2.3 L2: Risk Rule Aggregation and Update

L2 groups cards with similar z_i , risk type, and action semantics into pattern set $\mathcal{I}_j$. The aggregation map $\mathcal{A}$ derives rule Q_j from their triggers, recommended and prohibited actions, mitigations, and source IDs. Let n_j^+ , n_j^- , and n_j^0 count mitigation successes, risk failures, and inconclusive aborts. We define θ_j as the probability that a new observation provides decisive evidence for rule Q_j . With a Beta(α, β) prior, its posterior mean c_j is

$$\begin{aligned} (Q_j, c_j) &= \mathcal{A}(\{M_i\}_{i \in \mathcal{I}_j}), \\ c_j = \mathbb{E}[\theta_j \mid \mathcal{I}_j] &= \frac{n_j^+ + n_j^- + \alpha}{n_j^+ + n_j^- + n_j^0 + \alpha + \beta}. \end{aligned} \quad (2)$$

Here c_j measures decisive empirical support, not hazard probability. The source set of Q_j determines the counts in c_j and is checked before the rule becomes *active*. When a new L0 outcome supports or contradicts the rule, its IDs identify the associated cards, allowing the system to recompute support, revise the trigger, or mark the rule *retired*. Thus, the versioned chain $Q_j \rightarrow M_i \rightarrow E_i$ governs rule validation and maintenance rather than serving only as metadata.

At runtime, the rule gate admits actions recommended by every matched *active* rule and prohibited by none. An empty intersection stops execution and requests human review.

3.3 Role-Conditioned Risk Evidence Retrieval

Our second mechanism conditions evidence allocation on decision responsibility. For the same state and memory, the

Advocate prioritizes feasible progress and successful mitigation, the Critic prioritizes failures and stopping conditions, and the Decision agent prioritizes evidence for balancing task utility and risk. The novelty lies in applying the role condition inside retrieval, rather than using roles only to interpret a common evidence set. Given $\mathcal{X}_t$, each card yields $\phi_i = [\text{sim}, \text{sev}, \text{rel}, \text{rec}, \text{fit}] \in [0, 1]^5$, describing contextual similarity, severity, reliability, recency, and role fit. Role-specific weights $w^{(r)}$ convert these common features into responsibility-specific relevance.

For role r , $\mathcal{B}$ denotes a candidate evidence bundle, K limits its card count, and B_r is its token budget. The redundancy measure $d(i, j)$ discourages repeated risk types, outcomes, and mitigations; $\hat{\tau}(M_i)$ estimates the token cost of card M_i , and λ controls the redundancy penalty. Retrieval then selects

$$\begin{aligned} \mathcal{B}_r^* = \arg \max_{\mathcal{B} \subseteq \mathcal{M}_t} & \sum_{M_i \in \mathcal{B}} (w^{(r)})^\top \phi_i - \lambda \sum_{\substack{M_i, M_j \in \mathcal{B} \\ i < j}} d(i, j) \\ \text{s.t. } |\mathcal{B}| \leq K, & \sum_{M_i \in \mathcal{B}} \hat{\tau}(M_i) \leq B_r. \end{aligned} \quad (3)$$

For (sim, sev, rel, rec, fit), we use

$$\begin{aligned} w^{(\text{adv})} &= (0.35, 0.05, 0.20, 0.05, 0.35), \\ w^{(\text{crit})} &= (0.30, 0.25, 0.15, 0.05, 0.25), \\ w^{(\text{dec})} &= (0.35, 0.20, 0.20, 0.10, 0.15). \end{aligned}$$

Algorithm 1 greedily approximates Eq. (3); the experiments evaluate the role-specific weighting.

Each bundle stores its role and retrieval version, selected L1 cards and matched L2 rules, scores, rationales, budget use, and source IDs. A matched rule constrains the feasible actions, while its source set records which cases support that constraint. Agents declare the evidence used through `cited_experience_ids` and `cited_rule_ids`; IDs absent from the assigned bundle are rejected. The accepted

IDs are retained with the optimizer and Guard outputs, linking an executed or rejected action to the rule and cases that informed it.

Let $\mathcal{D}_t$ denote the three role reports and $\mathcal{F}_t$ the actions that satisfy hard constraints and matched L2 prohibitions. For action a , $U(a)$ is task utility, $R(a \mid \mathcal{B}_{\text{dec}}^*)$ is the combined scene-memory risk, and $Z(a \mid \mathcal{D}_t, \mathcal{B}_{\text{dec}}^*)$ is uncertainty from observation confidence, memory coverage, and role disagreement. The thresholds ρ and η bound these relative scores. The deterministic optimizer selects

$$\begin{aligned} a_t^* &= \arg \max_{a \in \mathcal{F}_t} U(a), \\ R(a \mid \mathcal{B}_{\text{dec}}^*) &\leq \rho, \\ Z(a \mid \mathcal{D}_t, \mathcal{B}_{\text{dec}}^*) &\leq \eta. \end{aligned} \quad (4)$$

Rule conflict, invalid output, an empty feasible set, or Guard rejection triggers `safe_stop`, `observe_again`, or human review. This deterministic boundary supports the two memory mechanisms by preventing semantic recommendations from bypassing action constraints.

Algorithm 1: Role-Conditioned Risk Retrieval

Require: State $\mathcal{X}_t$, roles $\mathcal{R}$, L1 cards $\mathcal{M}$, active L2 rules $\mathcal{Q}$, budgets $\{B_r\}$, limit K
Ensure: Role-specific evidence bundles $\{\mathcal{B}_r\}_{r \in \mathcal{R}}$

- 1: Validate $\mathcal{X}_t$, $\mathcal{R}$, $\mathcal{M}$, and $\mathcal{Q}$
- 2: **for** each role $r \in \mathcal{R}$ **do**
- 3: Compute all ϕ_i ; initialize candidates $\mathcal{C}_r$ and $\mathcal{B}_r \leftarrow \emptyset$
- 4: **while** $\mathcal{C}_r \neq \emptyset$ and $|\mathcal{B}_r| < K$ **do**
- 5: Select M_{i^*} by marginal gain in (3)
- 6: Remove M_{i^*} from $\mathcal{C}_r$
- 7: **if** $\hat{\tau}(\mathcal{B}_r) + \hat{\tau}(M_{i^*}) \leq B_r$ **then**
- 8: $\mathcal{B}_r \leftarrow \mathcal{B}_r \cup \{M_{i^*}\}$
- 9: **end if**
- 10: **end while**
- 11: Append matched active L2 rules, scores, rationales, and source IDs to $\mathcal{B}_r$
- 12: **end for**
- 13: **return** $\{\mathcal{B}_r\}_{r \in \mathcal{R}}$